

Plotting Wavefields from Research Module

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What have I been working on?

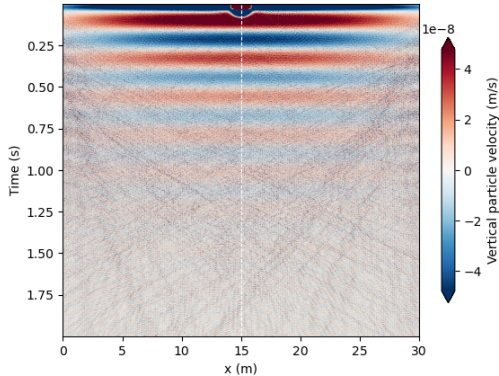
- Writing introduction
- Implementing feedback for introduction
- Setup of remote machine
- Running simulations with lower source frequency
- Plotting waterfall plots, wiggle plots, single traces and frequency content

Simulation parameters used

- 30m by 3m domain
- one simulation with 150m by 3m domain
- 3 layers, all layers 1m thick
- absorbing boundaries at edges of domain
- 10Hz vector point source
- moment tensor source for skier ($m_{yy}=0$, $m_{xx}=0$, $m_{xy}=3e4$)

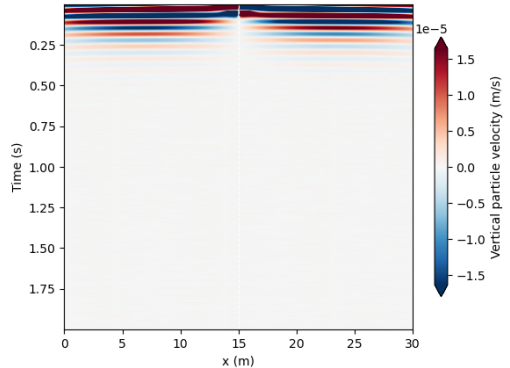
Wavefield Waterfall plots

Seismic waterfall plot below source location at 1m depth in snow



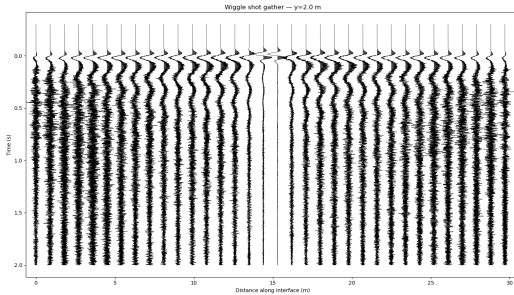
(a) Vector point 2D source with $f_y = -1$. Source location indicated by white line. Colorbar shows particle velocity (m/s).

Seismic waterfall plot below source location at 1m depth in snow

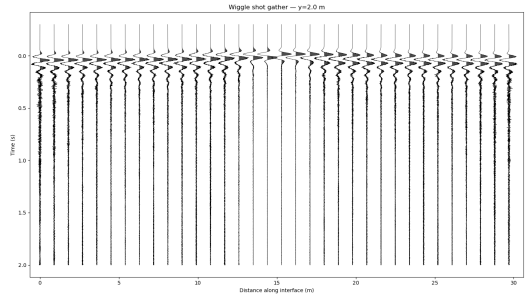


(b) Moment tensor source with $m_{xx} = 0$, $m_{yy} = 0$, $m_{xy} = 3 \times 10^4$. Source location indicated by white line. Colorbar shows particle velocity (m/s).

Wiggle Plots



(a) Wiggle plot for vector point source.



(b) Wiggle plot for moment tensor source.

Single traces

Problems / Questions

- Simulations take very long to run on remote machine (2+ hours)
- Does it make sense to add topography to simulations?
- Does it make sense to add eg old snow layer (make the simulations multilayered)?



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